ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70, seven questions of 10): RC circuits and lamp dimmers, transformer ratios, efficiency and cells in parallel, multivibrators and Schmitt triggers, bridge rectifiers and replacing a regulator, square-wave oscilloscope readings and probes, truth tables, Karnaugh maps and NAND logic, and three-phase induction motors. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 In a purely capacitive AC circuit, the current and the voltage are …

A in phase.
B 90° apart, with the current leading.
C 90° apart, with the voltage leading.
D 180° apart.

1.2 The unit of capacitance is the …

A henry.
B ohm.
C siemens.
D farad.

1.3 A bridge rectifier uses … diodes.

A four
B two
C one
D six

1.4 The output pulse time of a 555 timer in monostable mode is …

A 0,693RC.
B 1,44RC.
C 1,1RC.
D 2RC.

1.5 A transformer with more turns on the secondary than on the primary is a/an … transformer.

A isolation
B step-down
C step-up
D current

1.6 The output of a NOR gate is 1 only when …

A all the inputs are 0.
B all the inputs are 1.
C any input is 1.
D the inputs are different.

1.7 The speed of the rotating magnetic field in an induction motor is called the …

A rotor speed.
B slip speed.
C rated speed.
D synchronous speed.

1.8 A ×10 oscilloscope probe …

A doubles the signal.
B divides the signal by 10.
C multiplies the frequency by 10.
D removes the DC part of the signal.

1.9 The ripple on the output of a power supply is reduced by a larger …

A filter capacitor.
B series resistor.
C fuse.
D transformer core.

1.10 A Schmitt trigger is used to …

A amplify sound signals.
B store one bit of data.
C turn slow or noisy signals into clean square pulses.
D rectify AC.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 Identical cells connected in parallel can supply a larger current than one cell.

2.2 A square wave that is high for 2 ms and low for 2 ms has a duty cycle of 25%.

2.3 A bistable multivibrator has two stable states.

2.4 The iron (core) losses of a transformer change greatly when the load current changes.

2.5 The slip of an induction motor is zero at full load.

2.6 A NAND gate followed by a NOT gate works as an AND gate.

2.7 The output of a full-wave rectifier on a 50 Hz supply has a ripple frequency of 100 Hz.

2.8 In a series RC circuit, the same current flows through the resistor and the capacitor.

2.9 A stepper motor needs a commutator and brushes.

2.10 A fully charged capacitor blocks direct current.

(10)

[10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		**COLUMN B**
3.1	Ratio of the primary turns to the secondary turns of a transformer	A	slip
3.2	Percentage of one period during which a pulse is high	B	duty cycle
3.3	Difference between the synchronous speed and the rotor speed, as a fraction of the synchronous speed	C	bistable
3.4	Small AC variation left on the DC output of a power supply	D	phase angle
3.5	Circuit that stays in either state until it is triggered	E	ripple
3.6	Ratio of the output power to the input power	F	turns ratio
3.7	Two-terminal device used to trigger a TRIAC	G	Karnaugh map
3.8	Speed of the rotating magnetic field of an AC motor	H	DIAC
3.9	Angle by which the current leads or lags the voltage	I	synchronous speed
3.10	Grid used to group 1s and simplify a logic expression	J	efficiency

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 A 10 µF capacitor is connected in series with a 220 Ω resistor across a 230 V, 50 Hz supply. Calculate the:

4.1.1 capacitive reactance (2)

4.1.2 impedance (3)

4.1.3 current (2)

4.2 Explain how a DIAC and a TRIAC work together in a lamp dimmer. (3)

[10]

QUESTION 5

5.1 A transformer has 1 150 primary turns and 115 secondary turns. The primary is connected to 230 V. Calculate the secondary voltage. (2)

5.2 The transformer in QUESTION 5.1 delivers 2 A to its load and takes 50 W from the supply. Calculate its efficiency. (3)

5.3 Four cells, each with an emf of 1,2 V and an internal resistance of 0,2 Ω, are connected in parallel across a 1 Ω load. Calculate the total emf, the total internal resistance and the load current. (5)

[10]

QUESTION 6

6.1 Explain the difference between monostable, astable and bistable multivibrators. (6)

6.2 A 555 timer in monostable mode has R = 100 kΩ and C = 22 µF. Calculate the length of the output pulse. (3)

6.3 Give ONE use of a Schmitt trigger in a robot. (1)

[10]

QUESTION 7

7.1 Draw the circuit diagram of a bridge rectifier with a filter capacitor and a load resistor. (4)

7.2 Explain why the ripple becomes smaller when a bigger filter capacitor is used. (2)

7.3 The 7805 regulator on a robot's control board has failed. Describe FOUR steps to replace it safely and correctly. (4)

[10]

QUESTION 8

An oscilloscope is set to 5 V/div and 0,2 ms/div. A square wave on the screen rises 2 divisions above the 0 V line. In each cycle it is high for 1,5 divisions and low for 2,5 divisions. Calculate the:

8.1 peak voltage (2)

8.2 period (2)

8.3 frequency (2)

8.4 duty cycle (2)

8.5 Explain what a ×10 probe does, and why this must be taken into account when the voltage is read. (2)

[10]

QUESTION 9

9.1 Draw the truth table for $F = A \cdot B + C$. (4)

9.2 Use a Karnaugh map to simplify $F = \overline{A} \cdot \overline{B} \cdot C + \overline{A} \cdot B \cdot C + A \cdot \overline{B} \cdot C + A \cdot B \cdot C$. (3)

9.3 Show, with diagrams, how a NOT gate and an AND gate are made from NAND gates only. (3)

[10]

QUESTION 10

10.1 A 4-pole, three-phase induction motor is connected to a 50 Hz supply and runs at 1 440 r/min. Calculate the synchronous speed and the slip. (4)

10.2 Explain how the rotating magnetic field makes the rotor of an induction motor turn. (3)

10.3 Give THREE reasons why induction motors are used to drive conveyors instead of DC motors. (3)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Capacitive reactance: $X_C = 1 \div (2\pi fC)$	Impedance: $Z = \sqrt{(R^2 + X^2)}$; $I = V \div Z$
Transformer: $V_1 \div V_2 = N_1 \div N_2$	Efficiency: $\eta = (P_{out} \div P_{in}) \times 100\%$; $P = V \times I$
Cells in parallel: $E_T = E$; $r_T = r \div n$	$I = E_T \div (R + r_T)$
555 monostable: $t = 1{,}1 \times R \times C$	Frequency: $f = 1 \div T$; duty cycle = $(t_{high} \div T) \times 100\%$
Synchronous speed: $n_s = (120 \times f) \div p$ (p = number of poles)	Slip: $s = ((n_s - n) \div n_s) \times 100\%$